

From Single-User SC Trellis Decoding to a Two-User MAC Successive Cancellation Trellis (SCT) Decoder

Polar Codes for MAC Project — Implementation Notes

May 19, 2026

Abstract

We summarise the single-user successive cancellation (SC) trellis decoder of Wang, Honda, Yamamoto, Liu and Hou (ITW 2015) for polar codes over finite-state channels with memory, and then derive its two-user multiple-access channel (MAC) extension by composing it with the monotone chain rule decomposition of Öney (ISIT 2013). The two-user version is the *SCT decoder* used as the analytical baseline in our project. We state the recursive formulas, give a one-paragraph justification of why they are correct, and describe how the algorithm is realised in our repository.

1 Setting

A discrete finite-state channel is specified by

$$W(y_i, s_i \mid x_i, s_{i-1}),$$

where $x_i \in \mathcal{X} = \{0, 1\}$ is the input, $y_i \in \mathcal{Y}$ the output, and $s_i \in \mathcal{S}$ a finite hidden state evolving in time. The state at time 0 is known (say s_0). Memoryless channels are the special case $|\mathcal{S}| = 1$.

Polar transform. Let $\mathbf{G}_N = G^{\otimes n}$ with $G = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, $N = 2^n$. The input is $X_1^N = U_1^N \mathbf{G}_N$; SC decoding aims to recover U_1^N from Y_1^N . Savaş (2011) showed that the polar transform polarises finite-state channels in exactly the same way it polarises B-DMCs. What is needed in practice is a *decoder* that runs in time polynomial in $|\mathcal{S}|$ and N .

2 Single-User SCT (Wang/Honda/Yamamoto/Liu/Hou, ITW 2015)

2.1 Augmented joint probability

For memoryless channels the SC decoder uses

$$W_N^{(i)}(y_1^N, u_1^i) := P(Y_1^N = y_1^N, U_1^i = u_1^i),$$

which factorises along the polar tree. With memory this no longer factorises because the channel has correlated past and future. The paper's key idea is to add the *boundary states* s_0, s_N to the conditioning:

$$W_N^{(i)}(y_1^N, u_1^i, s_N \mid s_0) := P(Y_1^N = y_1^N, U_1^i = u_1^i, S_N = s_N \mid S_0 = s_0). \quad (1)$$

Conditioning on the entry state s_0 and pinning the exit state s_N makes the augmented quantity factor cleanly over a half-block, because everything before s_N is conditionally independent of everything after it given s_N .

2.2 Recursive formulas

The paper proves (Theorem 2 in [1]) that for $N = 2^n$, $1 \leq i \leq N$,

$$W_{2N}^{(2i-1)}(y_1^{2N}, u_1^{2i-1}, s_{2N} | s_0) = \sum_{s_N} \sum_{u_{2i}} W_N^{(i)}(y_1^N, u_{1,o}^{2i} \oplus u_{1,e}^{2i}, s_N | s_0) \cdot W_N^{(i)}(y_{N+1}^{2N}, u_{1,e}^{2i}, s_{2N} | s_N), \quad (2)$$

$$W_{2N}^{(2i)}(y_1^{2N}, u_1^{2i}, s_{2N} | s_0) = \sum_{s_N} W_N^{(i)}(y_1^N, u_{1,o}^{2i} \oplus u_{1,e}^{2i}, s_N | s_0) \cdot W_N^{(i)}(y_{N+1}^{2N}, u_{1,e}^{2i}, s_{2N} | s_N). \quad (3)$$

Here $u_{1,o}^{2i}$ and $u_{1,e}^{2i}$ are the odd- and even-indexed sub-sequences of u_1^{2i} — the standard polar split.

2.3 Why the recursion is correct

The argument has three ingredients:

1. The polar transform splits u_1^{2i} into $(u_{1,o} \oplus u_{1,e})\mathbf{G}_N$ in the first half-block and $u_{1,e}\mathbf{G}_N$ in the second — exactly as for memoryless channels.
2. Conditional on s_N , the past (Y_1^N, X_1^N, s_0) is independent of the future $(Y_{N+1}^{2N}, X_{N+1}^{2N}, s_{2N})$. So the joint factorises, and the two half-blocks each contribute a single $W_N^{(i)}$ term.
3. The difference between the odd index $(2i-1)$ and the even index $(2i)$ is exactly Arikan's split: the odd case still has to marginalise u_{2i} , the even case has already absorbed it.

Bringing those together gives (2)–(3). All inner steps reduce to ordinary marginalisation.

2.4 Complexity

The recursion is the same polar tree as the memoryless case, but at each of the $\mathcal{O}(N \log N)$ steps it must sum over $|\mathcal{S}|$ choices of s_N and store a value for every $(s_0, s_N) \in \mathcal{S}^2$. Total cost $\mathcal{O}(|\mathcal{S}|^3 N \log N)$. For ISI with a 1-tap state, $|\mathcal{S}| = 2$, so this stays $\mathcal{O}(N \log N)$ up to constants.

2.5 Recovering the per-bit posterior

The decoder needs the posterior $P(U_i | U_1^{i-1}, Y_1^N)$. From $W_N^{(i)}$ the joint probability $P(Y_1^N, U_1^i)$ is obtained by averaging over the boundary states,

$$P(Y_1^N = y_1^N, U_1^i = u_1^i) = \sum_{s_0, s_N} W_N^{(i)}(y_1^N, u_1^i, s_N | s_0) P(S_0 = s_0), \quad (4)$$

and the SC decision rule is the usual $\hat{u}_i = \arg \max_{u_i} P(Y_1^N, U_1^{i-1} \hat{u}_1^{i-1}, u_i)$.

3 Lifting to the Two-User MAC

3.1 Two-user MAC setting

A two-user binary-input MAC with memory has transition law

$$W(z_i, s_i | x_i, y_i, s_{i-1}),$$

where x_i and y_i are the two users' inputs, z_i the joint output, and s_i the channel state. The state s_i now generally has to encode all information needed to predict the next output. For the ISI-MAC

$$Z_i = (1 - 2X_i) + (1 - 2Y_i) + h[(1 - 2X_{i-1}) + (1 - 2Y_{i-1})] + W_i, \quad W_i \sim \mathcal{N}(0, \sigma^2),$$

the natural state is $s_i = (X_{i-1}, Y_{i-1}) \in \{0, 1\}^2$, so $|\mathcal{S}| = 4$. Inputs are independent uniform: $P_X(0) = P_X(1) = P_Y(0) = P_Y(1) = \frac{1}{2}$.

3.2 Önay's monotone-chain decomposition (ISIT 2013)

Let U^N and V^N be the polar pre-codewords for the two users. Önay [2] expands $I(Z^N; U^N, V^N)$ along a *monotone chain rule path* $b^{2N} \in \{0, 1\}^{2N}$: a 0 at step k emits an extra U symbol, a 1 emits an extra V symbol. Each monotone path produces a sequence of *coordinate channels* $W_N^{(b_k, i, j)}$, $k = i + j$, that are single-user binary-input channels (eq. (8) of [2]):

$$W_N^{(b_k, i, j)}(z^N, s^{k-1} | s_k) = \begin{cases} W_N^{(0, i, j)}(z^N, u^{i-1}, v^j | u_i) & \text{if } b_k = 0, \\ W_N^{(1, i, j)}(z^N, u^i, v^{j-1} | v_j) & \text{if } b_k = 1, \end{cases} \quad (5)$$

and these polarise (capacity terms go to 0 or 1) for every monotone path. The decoder works through $k = 1, \dots, 2N$ deciding one bit at a step, of user-1 if $b_k = 0$ or user-2 if $b_k = 1$. In particular:

- $b^{2N} = 0^N 1^N$ is the *corner-rate Class C* path: decode all of U first treating V as i.i.d. noise, then all of V conditioned on the just-decoded $\hat{X} = \hat{U} \mathbf{G}_N$.
- Other paths give intermediate rate splits and reach the dominant face of the MAC capacity region with arbitrary precision.

Önay's recursive formulas (his eqs. (16)–(19)) are simply the same polar recursion as in the single-user case, but they propagate the four-type coordinate channels $W_N^{(b_k, i, j)}$ instead of one type.

3.3 Composition with the state-lattice trick

Both Önay's MAC SC and Wang/Honda et al.'s state-lattice SC use exactly the same polar split. The composition is therefore straightforward: augment Önay's coordinate channels with the boundary states (s_0, s_N) exactly as in (1). Define

$$W_N^{(b_k, i, j)}(z_1^N, u^{i-1} \text{ or } u^i, v^{j-1} \text{ or } v^j, s_N | u_i \text{ or } v_j, s_0), \quad (6)$$

keeping all four Önay types $b_k = 0$ vs $b_k = 1$, even vs odd index. The recursion for each type takes exactly the same shape as the single-user SCT recursion (2)–(3): a sum over the shared boundary state s_N inside the product of two half-block factors. For example, the analogue of (2) for an odd-index $b_k = 0$ (U) step is

$$\begin{aligned} & W_{2N}^{(0, 2i-1, 2j)}(z_1^{2N}, u_1^{2i-2}, v_1^{2j}, s_{2N} | u_{2i-1}, s_0) \\ &= \frac{1}{2^2} \sum_{s_N} \sum_{u_{2i}, v_{2j-1}, v_{2j}} W_N^{(0, i, j)}(z_1^N, \tilde{u}^{i-1}, \tilde{v}^j, s_N | \bar{u}_i, s_0) \\ & \quad \cdot W_N^{(0, i, j)}(z_{N+1}^{2N}, \bar{u}^{i-1}, \bar{v}^j, s_{2N} | \bar{u}_i, s_N), \end{aligned} \quad (7)$$

where $\tilde{u} = u_{1,o} \oplus u_{1,e}$, $\bar{u} = u_{1,e}$ and similarly for $\tilde{v}$, $\bar{v}$. Three other shape-equivalent identities cover the $(b_k, \text{odd/even}) \in \{0, 1\} \times \{\text{odd, even}\}$ cases; they are the natural composition of Öney's eqs. (16)–(19) with the state-summation of (2)–(3). No new information-theoretic argument is needed beyond:

1. the polar split applies to u and v separately (Öney), and
2. conditional on s_N , the past and future of the channel are independent (Wang/Honda).

3.4 Complexity

Each step of the recursion for the MAC version sums over $|\mathcal{S}|$ choices of s_N and stores a table indexed by (s_0, s_N) , exactly as single-user. Thus the cost remains $\mathcal{O}(|\mathcal{S}|^3 N \log N)$. For ISI-MAC, $|\mathcal{S}| = 4$ and the algorithm runs in $\mathcal{O}(64 N \log N) = \mathcal{O}(N \log N)$.

4 Corner-Rate Specialisation and Equivalent Two-Stage Form

For the corner-rate path $b^{2N} = 0^N 1^N$, all U steps come before any V step. The recursion (7) simplifies because, in the U phase, each MAC coordinate channel $W^{(0,i,j)}$ with $j = N$ *marginalises* V as uniform i.i.d.: the joint channel $W(z | x, y)$ is replaced by its V -averaged version $\tilde{W}(z | x) = \frac{1}{2} \sum_y W(z | x, y)$. Equivalently, the entire U phase can be implemented in two steps:

- (a) Compute the per-position marginal $\log P(z_t | x_t)$ from *joint* forward–backward on the $|\mathcal{S}|$ -state trellis with i.i.d. uniform Y . This produces a scalar LLR $\ell_t^{(u)}$ for each position.
- (b) Run *ordinary* Arikan SC on $\ell_1^{(u)}, \dots, \ell_N^{(u)}$ to decide $\hat{U}$.

For stage 2, condition on $\hat{X} = \hat{U} \mathbf{G}_N$: now both X_t and X_{t-1} are known, so the state collapses to $s_t = Y_{t-1}$, a 2-state chain. Run a 2-state FB to obtain a scalar LLR $\ell_t^{(v)}$ and ordinary Arikan SC for $\hat{V}$. This is the version used in the project; it is mathematically equivalent to (7) specialised to $b^{2N} = 0^N 1^N$.

5 Realisation in the Project

Two implementations follow the development above; both are validated to agree within Monte-Carlo noise across $N \in \{16, \dots, 1024\}$:

- `polar/decoder_trellis.py` (*joint MAC SCT*): builds the 4-state leaf tensor for ISI-MAC, runs a single **forward-backward** on the full $|\mathcal{S}| = 4$ lattice, and feeds the resulting $(N, 2, 2)$ per-position marginals into a Ren-style computational-graph two-user MAC SC tree. Supports any monotone path b^{2N} via `polar/decoder_interleaved.py`.
- `scripts/local_analysis/chained_sct_4state.py` (*4-state chained corner-rate*): the two-stage form of Section 4 above. Stage 1 uses the same 4-state FB but marginalises Y per position to obtain scalar U-LLRs, then runs Arikan SC. Stage 2 conditions on $\hat{X}$ and runs a 2-state Y_{t-1} FB followed by Arikan SC. This decoder is the primary analytical baseline we plot against the neural decoders.

Numerical sanity. At ISI-MAC with $h = 0.3$, $\text{SNR} = 6$ dB, corner-rate path, the 4-state chained SCT matches NPD within Monte-Carlo noise at every $N \in \{16, \dots, 1024\}$ once the MC genie-aided design uses $\geq 50,000$ trials per N (and $\geq 200,000$ at $N = 1024$ to resolve the long reliability tail).

6 Summary

- The single-user SCT of Wang/Honda/Yamamoto/Liu/Hou (2015) generalises Arikan SC to finite-state channels by adding the boundary states (s_0, s_N) to every polar-tree intermediate quantity, see (1). The polar-tree recursion is unchanged in form (2)–(3); only an extra sum over s_N is added at each step.
- Öney (2013) showed that any monotone chain rule expansion of $I(Z; U, V)$ gives a polar code reaching every point on the dominant face of the MAC capacity region. His decoder is the same polar-tree recursion as single-user SC, applied to the four coordinate channels.
- Composing the two is mechanical: take Öney’s recursions for the four coordinate-channel types, augment each by (s_0, s_N) and add a sum over s_N at every composition — exactly as in Wang/Honda et al. The result is the two-user MAC SCT used in the project, complexity $\mathcal{O}(|S|^3 N \log N)$.
- For the corner-rate path $b^{2N} = 0^N 1^N$, the algorithm specialises to the two-stage form “state-lattice FB $\rightarrow$ scalar LLR $\rightarrow$ Arikan SC”, which is what we implement and use as our analytical baseline against the NPD and NCG decoders.

References

- [1] R. Wang, J. Honda, H. Yamamoto, R. Liu and Y. Hou, “Construction of Polar Codes for Channels with Memory,” *Proc. IEEE Information Theory Workshop (ITW), Fall 2015*, pp. 187–191.
- [2] S. Öney, “Successive Cancellation Decoding of Polar Codes for the Two-User Binary-Input MAC,” *Proc. IEEE Int. Symp. Inform. Theory (ISIT)*, 2013, pp. 1122–1126.
- [3] E. Şaşoğlu, “Polarization in the Presence of Memory,” *Proc. IEEE ISIT*, 2011, pp. 189–193.